LIVE INSTRUCTOR PRESENTATION / COMPLETE USER-INPUT WALKTHROUGH

User Input and Full Project Presentation

Step-by-step guide for presenting the running system, starting at the login screen.

START FROM YOUR CURRENT SCREEN
Keep the open login page and running services as they are. Use separate browser tabs or profiles for each main-app role. Field Operator uses the separate panel at http://localhost:5173.

1. Prepare before speaking

- From the project root in a second PowerShell window, run `docker compose ps`. Confirm the services are running.
- If a service is unhealthy, inspect `docker compose logs --tail 80 backend worker edge-gateway ai-service`. Avoid restarting Hardhat during the live demo.
- The seeded accounts share `DEMO_SEED_PASSWORD` from the root `.env`. Retrieve it privately with the command below; do not screen-share the terminal or show the password.

```
(Get-Content .\.env | Where-Object { $_ -match '^DEMO_SEED_PASSWORD=' }) -replace '^DEMO_SEED_PASSWORD=', ''
```

2. Login reference

Role	Email	Role selection / page
Admin	admin@demo.synthetic.test	Admin / localhost:3000
Manufacturer	manufacturer@demo.synthetic.test	Manufacturer / localhost:3000
Distributor	distributor@demo.synthetic.test	Distributor / localhost:3000
Healthcare Professional	healthcare@demo.synthetic.test	Healthcare Professional / localhost:3000
Field Operator	fieldoperator@demo.synthetic.test	Field Operator / localhost:5173

OPENING STATEMENT
This is a synthetic software demo of a vaccine supply chain. I will enter fictional batch and sensor data, follow the resulting alerts and shipment views, and verify a batch QR. It is not a clinical or production system.

STEP 1 Admin - readiness, orders, policy, and oversight (2 minutes)

Sign in on the already-open main-app login page

- Enter admin@demo.synthetic.test, select Admin, enter the private demo password, and click Sign In.
- Open Local-system readiness. Point to API, MongoDB, MQTT, Redis, AI, local Hardhat chain/contract, and test-email capture status. The panel reports health status, not secrets.
- Show the batch and policy area. Select an existing batch and read its approved limits from the screen. The seeded example commonly uses 2-8 degrees C; call it a demo policy, not a medical standard.
- Show inventory, orders, forecast analytics, supply-planning preview, reports, audit history, alerts, and Batch Story/traceability. Forecasts and stock are synthetic demo data.
- In order review, explain that Admin reviews a Healthcare order and can approve/decline it, assign eligible released stock, and choose a Distributor. Approval creates the shipment.
- If using policy or alert-review controls, show the reason/permission requirement. Only a designated current reviewer can complete quality review and clear a hold; response actions require notes.

ADMIN FEATURES TO NAME

Readiness; role-scoped order review and stock allocation; approved policy configuration; alert acknowledgement/action/review; audit and traceability; forecasting and approval-only planning; synthetic feedback analytics. The user directory is read-only.

EXPECTED FRESH-SEED STATE

Seeded batches are not manufacturing-released, so planning can show zero eligible allocation. Explain the release guard instead of claiming a shipment was created.

STEP 2 Manufacturer - enter a batch and complete release stages (2-3 minutes)

- Open a separate main-app tab/profile. Sign in with manufacturer@demo.synthetic.test and choose Manufacturer.
- Show account-owned batches, batch history, production/report information, inventory certificates, shipment context, alerts, analytics, blockchain audit, and Batch Story.
- Choose Create New Batch or New batch. Use the form below only if you want to create a new synthetic record; choose a unique ID each time.

Input field	Synthetic example	Validation / meaning
Batch ID	INS-DEMO-YYYYMMDD-01	Required and unique; use a fresh suffix if it already exists.
Vaccine type / product	Synthetic demo vaccine	Required; use the same name later in the order.
Batch quantity	100 doses	Positive whole number.
Supplier	Synthetic Demo Supplier	Required.
Material type	Synthetic vial material	Required.
Material quantity / unit	100 / test units	Positive whole number and unit label.
Expiry date	A future calendar date	Required; stored at the start of that date in UTC.
Remarks	Synthetic instructor demo only	Optional; max 2,000 characters.

Submit and show the result

- Click Create batch and report once. Wait for the success message and registration hash; if the dashboard refresh fails, refresh the dashboard instead of resubmitting.
- Show the new batch, raw-material/production/batch/certificate/report records, expiry, quantity, release stage, and registration transaction hash. Ownership comes from the signed-in Manufacturer account.
- Complete Production Sign and Lab Sign as separate actions if the live demo is prepared for release. Batch creation by itself does not release stock.
- Show the batch's history and select Show batch QR when available. Keep the QR or verification URL for the Healthcare step.

ACCURATE WORDING

Certificates are project records, not clinical quality certificates. Batch registration or a transaction hash does not certify product safety. A new batch also needs an approved policy and release before it can support an eligible shipment.

STEP 3 Virtual IoT model - configure, tick, and track (2-3 minutes)

- Return to Admin or the owning Manufacturer. Open Virtual IoT simulator and refresh devices.
- Use a batch with an approved policy. Select an existing device, or create one with a new Device ID, a synthetic Vehicle ID, the exact Batch ID, and a Shipment ID only when it is actually linked.

Simulator input	Suggested demo value / behavior
Scenario	NORMAL first; optional DOOR_OPEN or GPS_JUMP later.
Fixed seed	INS-DEMO-01. Reusing a seed/configuration makes the synthetic sequence repeatable.
Start time	Current ISO time; the default is suitable for the live demo.
Tick interval	5,000 milliseconds for the presentation.
Temperature range	Use values consistent with the selected demo policy, for example min 2 / max 8. Generator range is not the policy itself.
Initial battery	100 percent (demo default).
Optional GPS route	At least two {latitude, longitude} points, each within valid geographic ranges.

Create device and show one sensor record

- Click Create virtual device, then click Tick once. This creates one reading immediately, even when the device is stopped. Start begins repeated interval ticks; it does not send an immediate first tick.
- Show temperature, humidity, battery, GPS context, chart/history, batch/shipment linkage, and saved route/map. Stop repeated ticks after the example.
- Optional scenarios: COMPRESSOR_DRIFT raises temperature gradually; SENSOR_FLATLINE holds sensor values; LINK_LOSS demonstrates gateway buffering; GPS_MOVEMENT follows a synthetic route; BATTERY_DRAIN lowers battery faster.

EXPLAIN THE TRACKING MODEL

The simulator emits synthetic readings. MQTT is preferred; the Edge Gateway validates/buffers/forwards readings; the backend persists them and evaluates the approved policy. Map coordinates and vehicle route are simulated saved fixes, not physical GPS or independent tracking.

Supported synthetic scenarios

Scenario	What to describe
NORMAL	Values follow the configured normal range and synthetic route.
DOOR_OPEN / HIGH_TEMPERATURE	Temperature goes above the configured range.
LOW_TEMPERATURE	Temperature goes below the configured range.
COMPRESSOR_DRIFT	Temperature rises gradually over ticks.
SENSOR_FLATLINE	Temperature/humidity stop changing after an initial sample.
GPS_JUMP / GPS_MOVEMENT	Synthetic location jump or route movement.
BATTERY_DRAIN	Battery decreases faster while other values remain normal.
LINK_LOSS	Gateway buffers readings and forwards after the simulated interruption.

STEP 4 Manual reading - enter data and follow the alert (2 minutes)

Permissions

Signed-in role	Manual entry form	CSV import
Admin	Yes	Yes
Manufacturer	No	No
Distributor	Yes	No
Healthcare	No	Yes
Field Operator	Yes, shipment-derived	No

- For the clearest demo, use Admin and open Manual / CSV telemetry. Copy a real Batch ID from the dashboard and confirm that its policy is approved.
- Use the example values below. If the selected policy maximum differs, set temperature above the maximum shown on screen; do not assume all batches have the same policy.

Manual field	Example input	Constraint / explanation
Batch ID	Exact ID copied from dashboard	Must identify a real accessible batch.
Vehicle ID	INS-TRUCK-01	Synthetic vehicle identifier.
Temperature	11 degrees C	Valid range -40 to 60; use only if above the selected approved demo limit.
Humidity / battery	58 percent / 92 percent	Each 0 to 100.
Latitude / longitude	9.9312 / 76.2673	Latitude -90 to 90; longitude -180 to 180; synthetic location.
Timestamp	Current-time default	Required; cannot be future-dated.
Operator note	Synthetic instructor demo reading; not clinical data	Required; explains why the manual measurement was entered.

Submit and follow the record

- Read back the form and click Submit reading once. A queued message means downstream workers still need time; wait, then refresh.
- In the alert/temperature view, locate the latest batch and timestamp. Show the Manual badge, measured value, policy limits, reason, actor/source, and any linked shipment/location.
- Open the chart/history and show the saved point. A valid in-range reading may create no alert. AI explanation is advisory; the approved policy alone determines policy alerts and holds.

Follow the review and audit path

- If the event offers response controls, an authorized Admin or owning Manufacturer can acknowledge it and record an operational action with a note. Show the actor and append-only audit history.
- If the event creates a quarantine hold, an in-range recovery reading does not clear it by itself. A currently designated reviewer must review fresh evidence and explicitly clear the hold with a separate reason.

WHERE THE READING GOES

The API authenticates the actor and validates the fields, then queues accepted telemetry. Workers persist it, evaluate the approved policy, create role-scoped notifications, add an AI advisory, and stage a digest for local-chain Merkle anchoring. The dashboard labels Manual/CSV/Simulated provenance.

STEP 5 CSV input, record correction, and Field Operator (2 minutes)

Demonstrate CSV as a second user-entered source

- Sign in as Admin or Healthcare. In Manual / CSV telemetry, choose Download CSV template and keep the header names exactly as supplied.

```
batchId,vehicleId,temperature,humidity,battery,lat,lon,timestamp,note
REAL_BATCH_ID,INS-TRUCK-CSV,5.2,58,92,9.9312,76.2673,2026-11-03T10:30:00.000Z,Synthetic CSV demo reading
```

- Replace REAL_BATCH_ID with the exact batch ID. Use a current or past ISO timestamp, numeric values within the manual-reading ranges, and a synthetic note. Optional columns are `shipmentId` and `deviceId`.
- Save as CSV and select Upload CSV (max 5,000 rows). Show accepted/rejected totals, per-row reason, and the CSV source badge. Valid rows enter the same telemetry path in timestamp order.

Append-only correction

- If demonstrating correction, use the authorized correction workflow as Admin, Distributor, or Field Operator, provide a reason, and show that a new reading references the original with a supersedes link.
- The original is not edited or deleted. Never correct data by editing MongoDB; that would break audit and evidence history.

Field Operator input

- Open http://localhost:5173, sign in as fieldoperator@demo.synthetic.test, select an assigned shipment, and show its shipment/sensor history, gauges, alert feed, and map.
- Use Record manual sensor reading. Batch, shipment, and vehicle are derived from the selected shipment. Enter temperature, humidity, battery, latitude, longitude, current timestamp, and a required operator note, then submit once.

> **FIELD ROLE SCOPE**
> Field Operator can enter shipment-linked synthetic readings but cannot create batches, approve release, allocate stock, or manage system-wide policy. Its records are still subject to shared validation and role-scoped access.

STEP 6 Distributor - assigned shipment, tracking, and handoff (2 minutes)

- Open a separate main-app tab/profile. Sign in with distributor@demo.synthetic.test and select Distributor.
- Open shipments and show assigned shipment IDs, batch/product/quantity, workflow status, route/map context, saved GPS/telemetry history, scoped alerts, shipment history, and report export.
- For an eligible shipment assigned to this Distributor, perform the available sequence: Verify batch -> Dispatch shipment -> Supply hospital. Refresh after each completed action and read the resulting status/event.
- If blocked, inspect the message. Common guards are release/sign-off, batch/order/product match, distributor assignment, remaining quantity, active hold, or wrong workflow state. Explain the guard; do not bypass it.
- Do not create a duplicate shipment when Admin has already allocated the hospital order. Only use a shipment assigned to this account.

What the tracking screen means

- The tracking/visualizer joins accessible shipment, batch, telemetry, location, and alert data. Status colors summarize saved workflow/alert state; the map and GPS labels are simulated.
- The truck position follows persisted shipment workflow progress in the visualizer; it is not a timer-based live vehicle location. The historical map plots saved synthetic fixes.
- An alert not visible to this Distributor can be correct when its shipment or organization is not assigned to this account.

TRANSITION TO HEALTHCARE

After a successful Supply hospital event, Healthcare can refresh Deliveries and confirm receipt. If no eligible shipment exists, show the Distributor page and explain that seeded batches have not passed release gates.

STEP 7 Healthcare - enter an order and confirm receipt (2-3 minutes)

- Open a separate main-app tab/profile. Sign in with healthcare@demo.synthetic.test and choose Healthcare Professional.
- Open Workflow and locate Hospital Vaccine Order. Hospital identity is read-only and comes from this account.

Order field	Synthetic example	Validation / meaning
Region	Demo District	Required text.
Vaccine type	Same product as eligible batch	Required and must match allocation stock.
Requested doses	25	Positive whole number.
Priority	Normal	Choose High, Normal, or Low.
Order notes	Instructor demo reference INS-ORD-01	Optional; synthetic reference helps find this order.

- Submit Submit Vaccine Order once. Show the order in the submitted-order list and its pending/review status.
- Switch to Admin. In order review, inspect the vaccine type and quantity; select an eligible released batch with sufficient remaining stock and the correct Distributor; approve or decline with the normal controls.
- After approval, switch to Distributor. Refresh Shipments and perform Verify batch, Dispatch shipment, and Supply hospital only when available and eligible.
- Return to Healthcare, refresh Deliveries, enter an optional synthetic receipt note, and select Confirm Received once. Show the receipt, history, and derived stock change.

STOCK INTEGRITY
A confirmed shipment receipt adds hospital stock. Do not type inventory to fake a delivery. The backend checks shipment state and quantity; seeded unreleased stock can correctly prevent order allocation.

STEP 8 Healthcare - administration, reports, and feedback (1-2 minutes)

Optional synthetic administration and report

- Only continue after a real confirmed demo receipt is visible. Open Record Vaccination & Report and select a confirmed batch from the list.
- Enter an invented patient/reference (for example `INS-REF-001`), doses used no greater than the available stock, a synthetic reporting authority, and optional notes.
- Click Record vaccination and report once. Show the resulting administration/report and reduced remaining stock. The form cannot add stock.
- Use only invented references. The app's patient/reference field is not a place for real patient information.

Optional feedback input

Feedback field	Demo input / rule
Feedback text	At least 10 characters, max 5,000; use fictional text such as 'Synthetic demo feedback for review.'
Category / severity	Choose Side Effects, Effectiveness, Storage Concern, Administration Issue, or Other; Low, Medium, or High.
Batch / observation date	Choose a visible batch; date must be valid and not in the future.
Age group / region	Optional demo context: Child, Adult, Senior; synthetic region.
Submitted by	Read-only account-derived value.
Analytics consent	Starts unchecked. Leave unchecked unless the presentation explicitly needs the demo consent feature.

- Submit once and show the sentiment label/score and saved feedback history. The default demo sentiment is a heuristic/advisory, not clinical analysis.
- Show Reports, stock history, feedback history, and export controls where time permits. Appointments may be read-only/empty; do not describe unavailable scheduling as working.

> **CSV PERMISSION REMINDER**
> Healthcare can import CSV readings through Manual / CSV telemetry, but cannot use the manual sensor form. Its orders, batches, shipments, alerts, and QR access remain account/organization scoped.

STEP 9 QR verification, model summary, and close (2 minutes)

Verify the Manufacturer QR in Healthcare

- From Manufacturer, open Show batch QR for an accessible batch. In Healthcare, open Verify batch QR.
- Use the camera if supported, or paste the verification URL. Entering only a batch ID checks batch identity and does not prove sensor history.
- Read two outcomes separately: batch identity; and sensor evidence status - anchored, pending, or unavailable/no proof. Wait about 30 seconds for the Compose Merkle window if pending, then refresh.
- The local Hardhat contract stores Merkle roots/metadata; verification checks the sensor commitment against the configured local chain and scoped backend record. It proves integrity of committed data, not physical measurement, vaccine quality, or regulatory compliance.

Working models and outputs

Model / feature	Input	Output and authority
Approved cold-chain policy	Batch-linked temperature/humidity and approved limits/duration	Authoritative policy alerts and holds; review/clear permissions are audited.
AI temperature advisory	Time-based temperature features and recent sensor history	Advisory score/explanation with 2-of-3 persistence; it does not control policy holds.
Humidity baseline	Humidity reading compared with explanatory baseline	Context metric only; not a learned safety classifier.
GPS plausibility	Saved coordinates and timestamp intervals	Separate speed-rule advisory; GPS values are simulated.
Regional demand forecast	Synthetic historical order series	SARIMA compared with seasonal-naive baseline; synthetic evaluation only.
Supply planning	Orders, inventory, expiry/capacity assumptions	Review-only recommendation; human approval required; no automatic dispatch.
Feedback sentiment	Synthetic free-text feedback	Demo heuristic label/score and Admin review; experimental BERT is disabled in Compose.
Merkle QR proof	Stored reading digest and time-window root	Local-chain integrity check when evidence exists; status can be pending/unavailable.

CLOSE WITH THIS SUMMARY

We followed synthetic user inputs from manufacturer batch creation and sensor readings through policy alerts, role-scoped tracking, order allocation, distributor handoff, Healthcare receipt, and QR proof. The AI and forecasts are advisory, all operational data is synthetic, and the chain is local.

Live-demo recovery

- Refresh the current dashboard and check `docker compose ps`. For logs run `docker compose logs --tail 80 backend worker edge-gateway ai-service`.
- If a view lacks a record, explain role/organization scope. If an operation is blocked, show the guard. If QR is pending, wait and refresh. Never edit database records to create a presentation result.
- Finish with `docker compose down` when done. Avoid `docker compose down -v`; the local Hardhat chain history is volatile and does not survive a restart.